WORKED SOLUTION

STEP I 1999, Question 12

Probability and Statistics • Loaded Dice Inequalities

STEP I 1999, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q12

This question has a full guided treatment in *Beyond STEP*, Section 9.13, *Loaded Dice*: https://beyondstep.keithmcnulty.org/ch09/loaded-dice.html

The key idea

The inequality in the first line is the whole engine of the question. If every total were equally likely, then the two "corner" totals 2 and 12 would each force a product of probabilities to be $\frac{1}{11}$, and the total 7 would then contain two cross terms whose sum has the form $c\,(x + x^{-1})$. The inequality makes that sum too big. Part (ii) is the same idea in a different guise: a sum of squares is smallest when everything is equal.

Solution

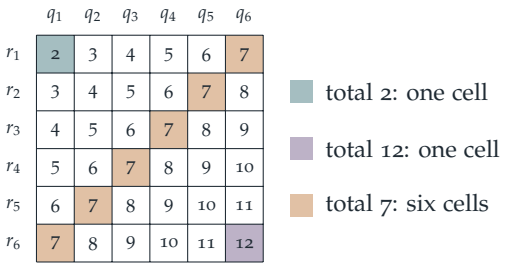
Part (i)

The inequality. For $x > 0$ we can take square roots, and

$$x + x^{-1} - 2 = \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 \geq 0,$$

so $x + x^{-1} \geq 2$, with equality only when $\sqrt{x} = 1/\sqrt{x}$, that is, $x = 1$. (Equivalently, multiply $(x-1)^2 \geq 0$ by the positive number x^{-1}.)

The dice. Let p_s be the probability that the total is s, for $2 \leq s \leq 12$. Since the dice are independent, the probability of the pair (first shows i, second shows j) is $q_i r_j$. The grid below shows which pairs give the totals we need.



The total 2 arises only from $(1,1)$ and the total 12 only from $(6,6)$, so

$$p_2 = q_1 r_1, \qquad p_{12} = q_6 r_6.$$

The total 7 arises from six pairs, including $(1,6)$ and $(6,1)$. All six terms are non-negative, so

$$p_7 = q_1 r_6 + q_2 r_5 + \cdots + q_6 r_1 \;\geq\; q_1 r_6 + q_6 r_1.$$

Now suppose, for contradiction, that all eleven totals are equally likely. They are exhaustive, so each has probability $\frac{1}{11}$. Then $q_1 r_1 = \frac{1}{11}$ and $q_6 r_6 = \frac{1}{11}$. In particular none of q_1, r_1, q_6, r_6 is zero, so we may write $r_1 = \dfrac{1}{11 q_1}$ and $r_6 = \dfrac{1}{11 q_6}$. Substituting,

$$\begin{aligned} p_7 \;\geq\; q_1 r_6 + q_6 r_1 &= \frac{q_1}{11 q_6} + \frac{q_6}{11 q_1} = \frac{1}{11}\left(x + x^{-1}\right), \qquad \text{where } x = \frac{q_1}{q_6} > 0, \\ &\geq \frac{2}{11} \end{aligned}$$

by the inequality we have just proved. But $p_7 = \frac{1}{11} < \frac{2}{11}$, a contradiction. So, however the two dice are loaded, the totals $2, 3, \ldots, 12$ cannot all be equally likely.

Ask yourself Why was it important to note that q_1 and q_6 are non-zero?

We divided by q_1 and q_6, and the inequality $x + x^{-1} \geq 2$ needs $x > 0$. Both are guaranteed because each product $q_1 r_1$ and $q_6 r_6$ equals $\frac{1}{11} \neq 0$. Without that step the argument would have a hole: a die that never shows 1 is perfectly legal in general.

Part (ii)

Let S be the event that the two throws of the first die give the same number. The throws are independent, so the probability that both show j is q_j^2, and these six events are disjoint. Hence

$$\mathrm{P}(S) = q_1^2 + q_2^2 + \cdots + q_6^2.$$

We want the smallest this can be, given $q_j \geq 0$ and $\sum q_j = 1$. Expand a sum of squares that measures how far the die is from fair:

$$0 \leq \sum_{j=1}^{6} \left(q_j - \tfrac{1}{6}\right)^2 = \sum_{j=1}^{6} q_j^2 - \tfrac{1}{3}\sum_{j=1}^{6} q_j + 6 \cdot \tfrac{1}{36}$$
$$= \sum_{j=1}^{6} q_j^2 - \tfrac{1}{3} + \tfrac{1}{6} = \mathrm{P}(S) - \tfrac{1}{6}.$$

So $\mathrm{P}(S) \geq \frac{1}{6}$, with equality exactly when every $q_j = \frac{1}{6}$, the fair die.

Ask yourself Where does $1/36$ come from, and why is it a red herring?

$\frac{1}{36}$ is the probability of one *particular* pair, such as a double six, with a fair die. "The numbers are the same" covers six such pairs, so for a fair die it has probability $6 \times \frac{1}{36} = \frac{1}{6}$, and loading the die can only increase it.

So the answer is **no**: whatever the values of q_j, the probability that the two numbers are the same is at least $\frac{1}{6}$, which is far more than $\frac{1}{36}$.

Checks

- Fair dice: $p_2 = p_{12} = \frac{1}{36}$ but $p_7 = \frac{6}{36}$, clearly unequal, as the result predicts.
- A numerical search over 20,000 random pairs of loaded dice never got all eleven totals within 0.05 of $\frac{1}{11}$, and exact checks with fractions confirmed $p_7 \geq q_1 r_6 + q_6 r_1$ every time.
- For the die with $(q_1, \ldots, q_6) = (0.1, 0.2, 0.3, 0.15, 0.15, 0.1)$ we get $\mathrm{P}(S) = 0.195 \geq \frac{1}{6}$; a simulation of 200,000 double throws gave 0.194.

Common slips

- Assuming the two dice are identical ($q_i = r_i$). The question allows them to be loaded differently, and the argument above doesn't need them to match.
- Writing $p_7 = q_1 r_6 + q_6 r_1$. There are four more terms; we only need an inequality, so throw them away, but say why ($q_i r_j \geq 0$).
- In part (ii), answering with $\sum q_j^2 \geq \frac{1}{36}$ or forgetting that "the same" means any of six outcomes.

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Part (ii) is the statement that $\sum q_j^2 \geq \frac{1}{n}\left(\sum q_j\right)^2$, a special case of the Cauchy–Schwarz inequality; the quantity $\sum q_j^2$ is called the *collision probability* and appears in information

theory and in the analysis of hashing. Part (i) is a classic result about generating functions: the total of two dice has probability generating function $Q(t)R(t)$, and a neat alternative proof notes that uniform totals would force $1 + t + \cdots + t^{10}$ to be a constant times a product of two real polynomials of degree 5, each of which has a real root, whereas $1 + t + \cdots + t^{10} = (t^{11} - 1)/(t - 1)$ has none. Generating functions of this kind are developed in Grimmett and Welsh, *Probability: An Introduction*.

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